

# Anjana Deivasigamani

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## Technical Skills

**AI & LLMs** : embeddings, FAISS, Hugging Face, LangChain, LLaMA, OpenAI API, pgvector, prompt engineering, RAG

**ML Libraries** : NLTK, PyTorch, scikit-learn, spaCy, TensorFlow, XGBoost

**Data Engineering** : Apache Airflow, BeautifulSoup, Docker, FastAPI, Pandas

**Data Validation & QA** : anomaly detection, data profiling, Great Expectations, JSON/metadata processing

**Cloud & Data Platforms** : GCP (BigQuery, Cloud Run, Cloud SQL, GCS), GitHub Actions

**Databases** : MySQL, PostgreSQL

**Dashboards & Reporting** : ggplot2, Matplotlib, Seaborn, Streamlit, Tableau

**Programming Languages** : Bash, Python, R, SQL

## Experience

### Beauty Intelligence LLC

Alabama, US

*Data Scientist Intern*

*July 2025 – Dec 2025*

- Eliminated ~4 hours of manual processing per cycle across 2M+ transactions per export by re-engineering bi-weekly Nielsen syndicated POS export validation, migrating from manual Power Query workflows to automated Python scripts orchestrated via Apache Airflow.
- Built validation logic to detect and resolve malformed UPCs, schema inconsistencies, and anomalous product descriptors, ensuring data completeness before downstream BigQuery ingestion.
- Detected SKU-level anomalies across 40GB+ of data spanning 300+ hair care CPG brands; applied FP-Growth market basket analysis to surface co-purchase patterns that informed marketing and promotional strategy.
- Built a BigQuery ingestion pipeline surfacing cross-brand affinity signals for 10+ hair care brands; replaced manual ad hoc reporting with self-serve Streamlit co-purchase dashboards.

### Mr. Cooper (Xome Pvt. Ltd.)

Chennai, IN

*Software Developer Intern*

*Jan 2023 – July 2023*

- Designed a relational data model in MS SQL Server for Xome's high-volume auction platform, supporting a network of 875,000+ registered bidders; defined indexing strategies and data integrity constraints that improved query/lookup performance and prevented duplicate bids and race conditions during concurrent transactions.
- Authored complex SQL queries and stored procedures for auction lifecycle operations, including bid tracking and transaction reconciliation supporting a platform with \$16B+ in cumulative gross sales.
- Performed root-cause analysis on reconciliation mismatches, resolving data discrepancies to ensure transactional accuracy.
- Built a full-stack auction management prototype in C# and ASP.NET Core MVC to validate end-to-end data flows against the relational data model, surfacing schema design issues prior to production deployment.

## Projects

### 🔗 SellerCentral Chatbot System | *LangChain, FAISS, Hugging Face, LLaMA, BigQuery, GCP, Docker, FastAPI, Airflow*

- Designed and deployed a production-grade RAG pipeline using LangChain and LLaMA via Groq; ran systematic experiments across Hugging Face embedding models, FAISS indexing strategies, and chunking configurations; benchmarked against baseline LLM approaches using Precision@k, Recall@k, BLEU, and ROUGE-L, achieving 95% response accuracy and sub-2-second latency for 90% of queries.
- Established a data quality framework with automated anomaly detection and bias checks across hundreds of thousands of records, flagging 27% as anomalous; enforced data integrity with 7 test modules covering schema validation, DAG execution, and ETL correctness to ensure semantic search reliability.
- Architected an Airflow ETL pipeline ingesting Amazon review data into BigQuery with incremental loading across 4 product categories; deployed end-to-end as a Dockerized FastAPI service on GCP Cloud Run with CI/CD via GitHub Actions.

## Education

### Northeastern University, Boston, MA

**GPA: 3.67/4.0**

*Master of Science – Data Science*

*May 2026*

### Anna University, Chennai, IN

**GPA: 8.49/10.0**

*Bachelor of Engineering – Computer Science and Engineering*

*May 2023*

## Publications

“Customer Emotion Analysis on Food Review Images Using Deep Learning: A Review.” *Disruptive Technologies for Sustainable Development – Proceedings of ICARSI-23*, pp. 208–213. CRC Press / Taylor & Francis, 2024. DOI: [10.1201/9781003428473-44](https://doi.org/10.1201/9781003428473-44)